TECHNOLOGICAL UNIVERSITY OF THE PHILIPPINES
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Computer Programming 3
Documentation

(TransactionFlow / ReportGeneration)

Carcao Pizza Express

PizzaStorePoint-of-SaleSystem

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INTRODUCTION

DISCUSSION

Carcao Pizza Express is a pizza store point-of-sale (POS) system designed to streamline how the store handles daily operations — from taking customer orders and processing payments to tracking ingredient stock and generating sales reports. Instead of relying on manual order slips and handwritten inventory counts, the system brings product management, ingredient and recipe tracking, supplier records, transactions, expenses, and reporting into a single application.

The project was developed by Carlo S. Isidro and Lance Mathew Dumagcao as part of the Computer Programming 3 requirements, focusing on transaction flow and report generation.

The core concept of the system is to connect every stage of a sale — from selecting menu items on the order screen, deducting the exact ingredient quantities used per recipe, updating stock and expiry batches, to logging the completed transaction and its receipt — so that inventory levels, expenses, and income figures stay accurate and up to date without manual reconciliation.

DATABASE SCHEMA

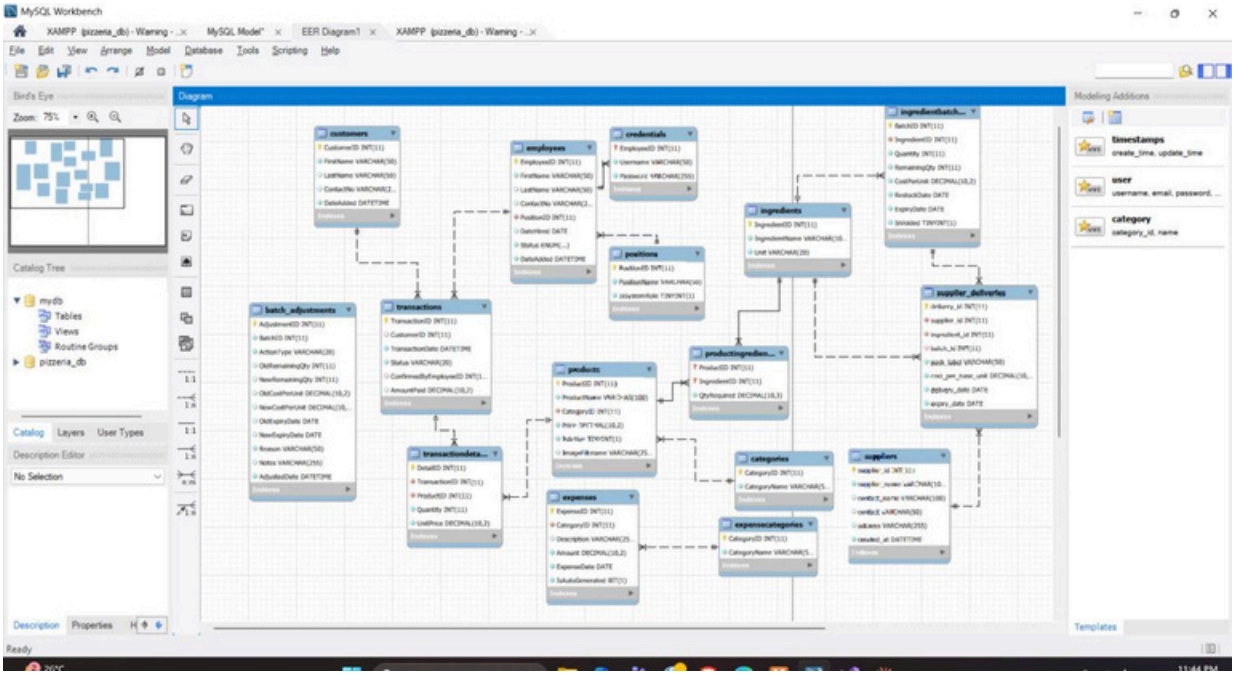


Figure 1. Entity-Relationship Diagram of the Carcao Pizza Express database (pizzeria_db)

The schema is built around a sales-to-stock relationship: every product sold is linked toarecipe made up of ingredients, and every ingredient is tracked down to individual restock batches with their own costand expiry date. When a transaction is completed, the system looks up each product's recipe in productingredientsand deducts the corresponding quantity from the matching ingredientbatch records, which is what keeps the Stocksand Expiry Deletion tabs accurate in real time. Supplier deliveries, batch adjustments, and auto-generated restock expenses all reference back to this same batch, giving the store a full audit trail from purchase to sale.

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TABLE DESCRIPTIONS

customers
Stores customer records (first name, last name, contact number, date added) used when placing and looking up orders.

employees
Storesstaffinformation including name, contact number, position, date hired, and active/inactive status.

credentials
Storestheusername and password of each employee, linked to their employee record, for logging into the system.

positions
Stores jobpositions (e.g., Manager, Cashier, Cook, Waiter, Janitor, Dishwasher) and whether a position has system access.

categories
Storesproduct classifications such as Pizza, Drinks, and Side Dish.

products
Storesmenuitems sold by the store, including name, category, price, active status, and image filename.

ingredients
Storesthemasterlist of raw ingredients (name and unit of measure, e.g., g, ml, pcs) used across recipes.

ingredientbatch
Recordsindividualrestock batches per ingredient, tracking quantity, remaining quantity, cost per unit, restock date, and expiry date.Thisis the basis for the Stocks and Expiry Deletion tabs.

productingredients
The recipetable —links each product to the ingredients and quantities required to make it, so the system knows what to deduct fromstock per sale.

suppliers
Storessupplier information including contact person, contact number, and address.

supplier_deliveries
Recordsdeliveriesreceived from suppliers, including the ingredient, batch, pack size, purchase cost, delivery date, and expiry date.

transactions
Stores thesummary of each sales transaction, including customer, date, status (Pending, Completed, Voided), and amount paid.

transactiondetails
Storestheindividual product line items included in each transaction, with quantity and unit price, used to compute the subtotal per item.

batch_adjustments
Records changesmade to an ingredient batch (restocks, corrections, or expired-batch deletions), keeping an audit trail of old and newquantities, cost, and expiry dates with a reason.

expensecategories
Storesexpenseclassifications such as Utilities and Restock.

expenses
Storesoperational expenses, including manually entered expenses and expenses auto-generated whenever ingredients are restocked.

WELCOME PAGE



Figure 1. Welcome page of the Carcao Pizza Express ordering system.

Carcao Pizza Express is a pizza-on-the-go ordering platform that lets customers log in, browse the menu, and place orders in just a few clicks. The welcome page serves as the entry point to the system, greeting customers with the brand's signature look and guiding them straight to the Login and Order actions. From here, every order flows into the store's backend, where it's tracked against real-time stock and ingredient batches — ensuring that every slice served is backed by fresh, available ingredients.

TRANSACTION: MAJOR FUNCTIONALITIES

The system defines two (2) levels of access, each with its own set of functions.

1. Administrator Functions

The administrator logs in through the login form and lands on the Admin Dashboard, which gives full access to every module of the system.

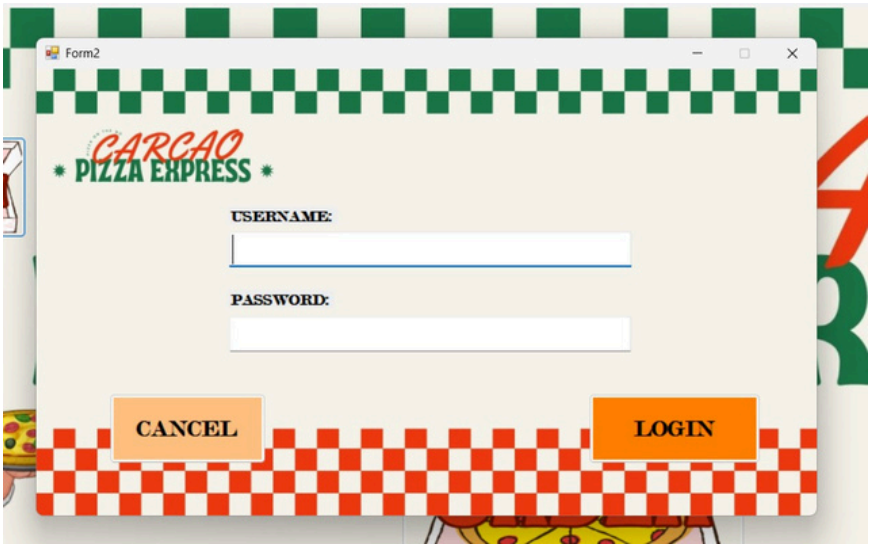


Figure 2. Login form

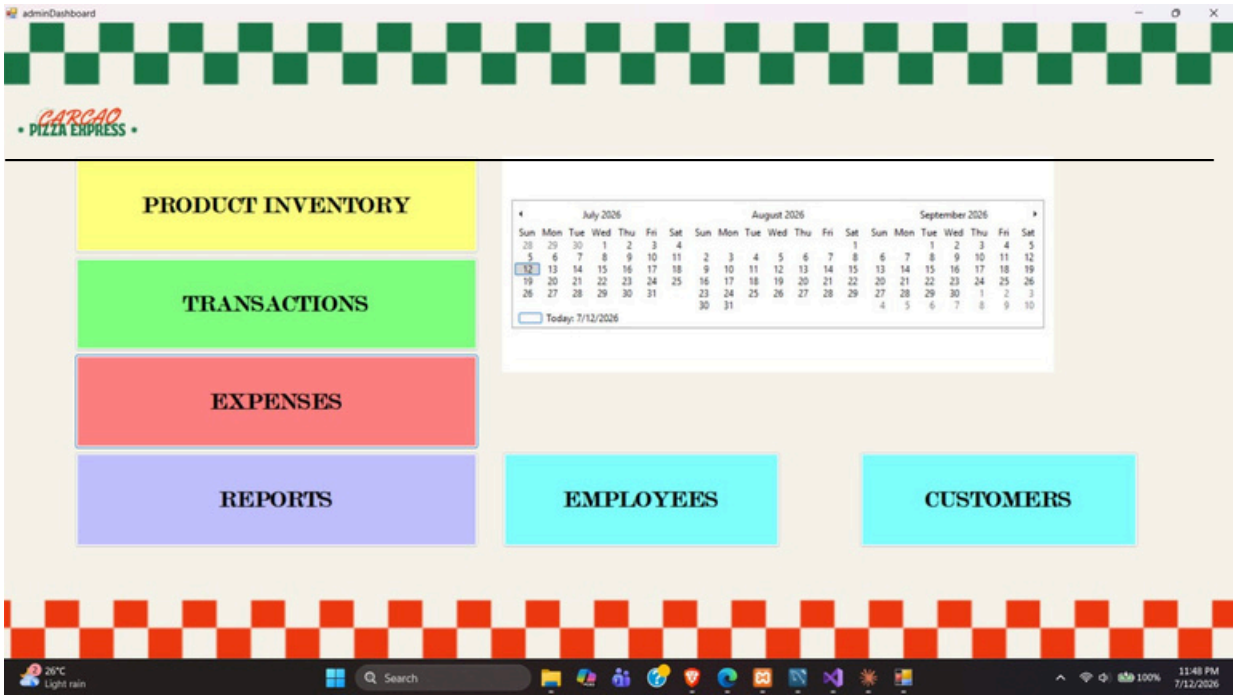


Figure 3. Admin Dashboard — Product Inventory, Transactions, Expenses, Reports, Employees, and Customers

- Manage product inventory — add, update, delete, and view products, each with a category, price, producible quantity, and photo (Products Menu tab).

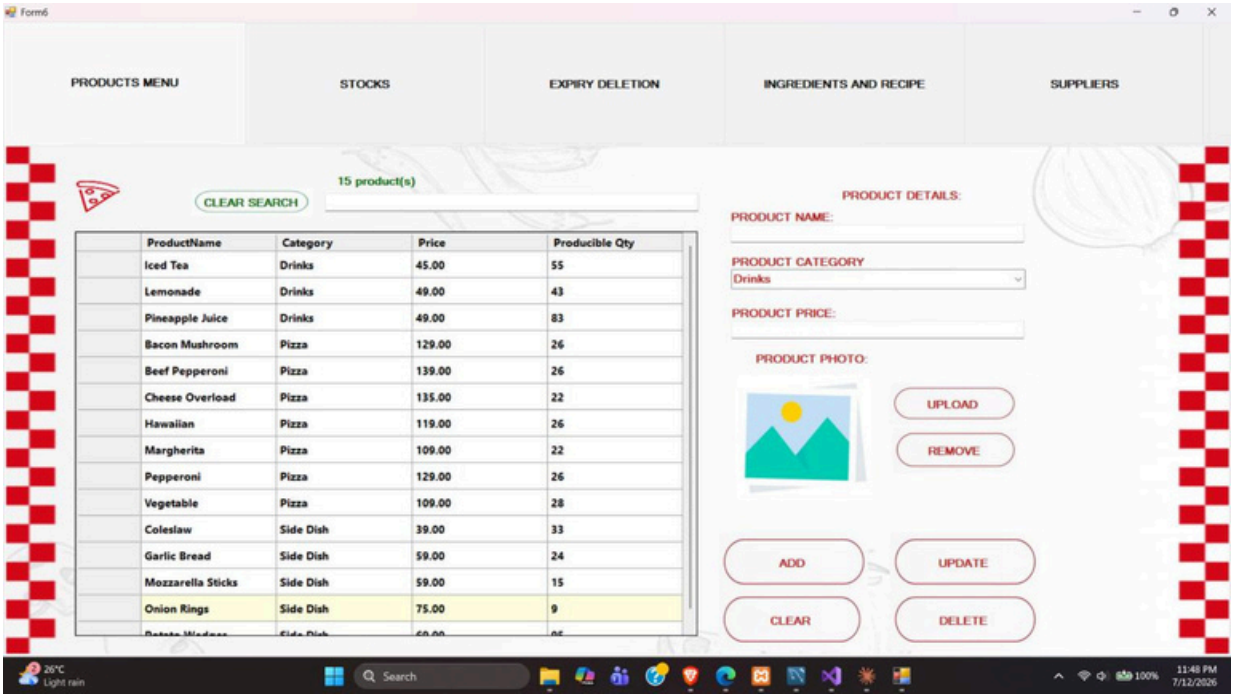


Figure 4. Product Inventory — Products Menu tab

- Manage stock batches — restock ingredients, record batch quantity/cost/expiry, and log manual adjustments (Stocks tab).

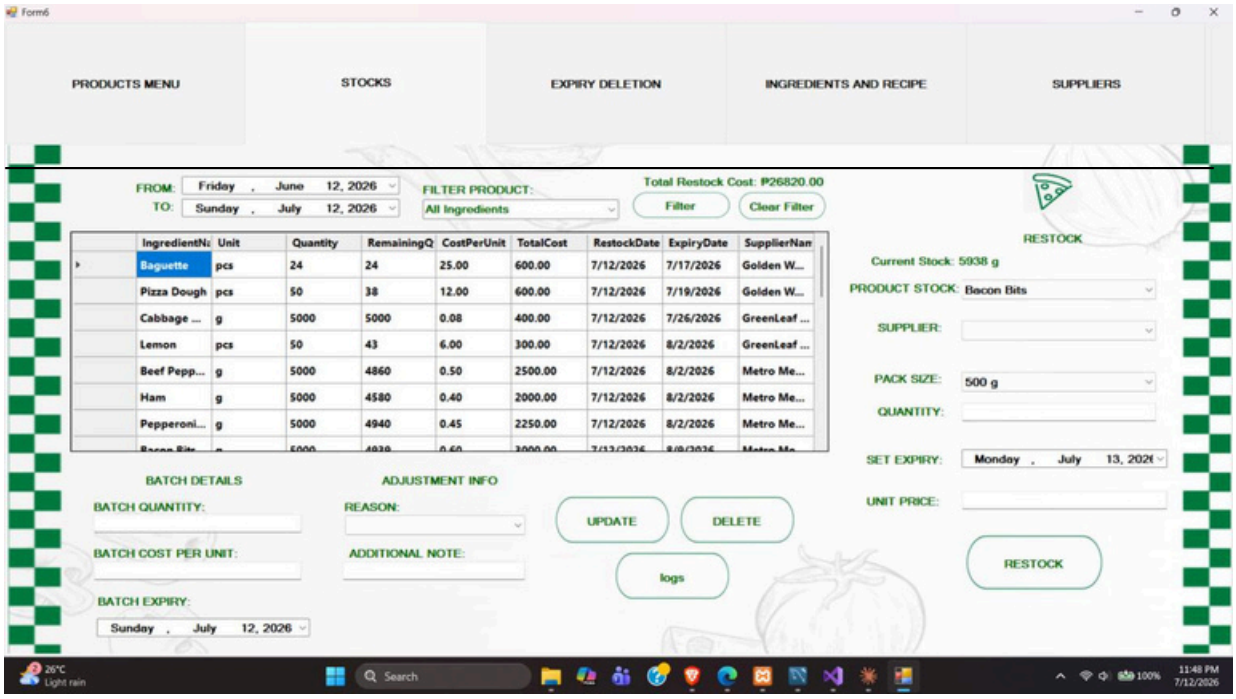


Figure 5. Product Inventory — Stocks tab

- Detect and remove expired ingredient batches (Expiry Deletion tab).

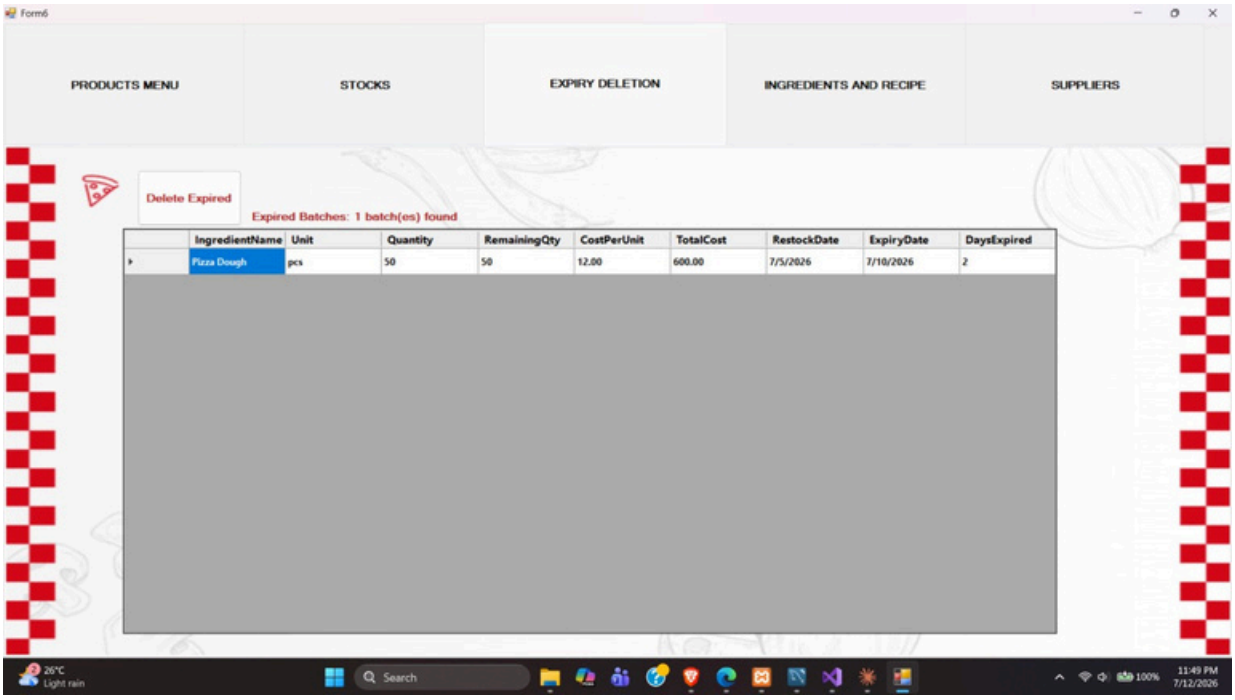


Figure 6. Product Inventory — Expiry Deletion tab

- Manage ingredients and build/edit product recipes by linking ingredientsand required quantities to each product (Ingredients and Recipe tab).

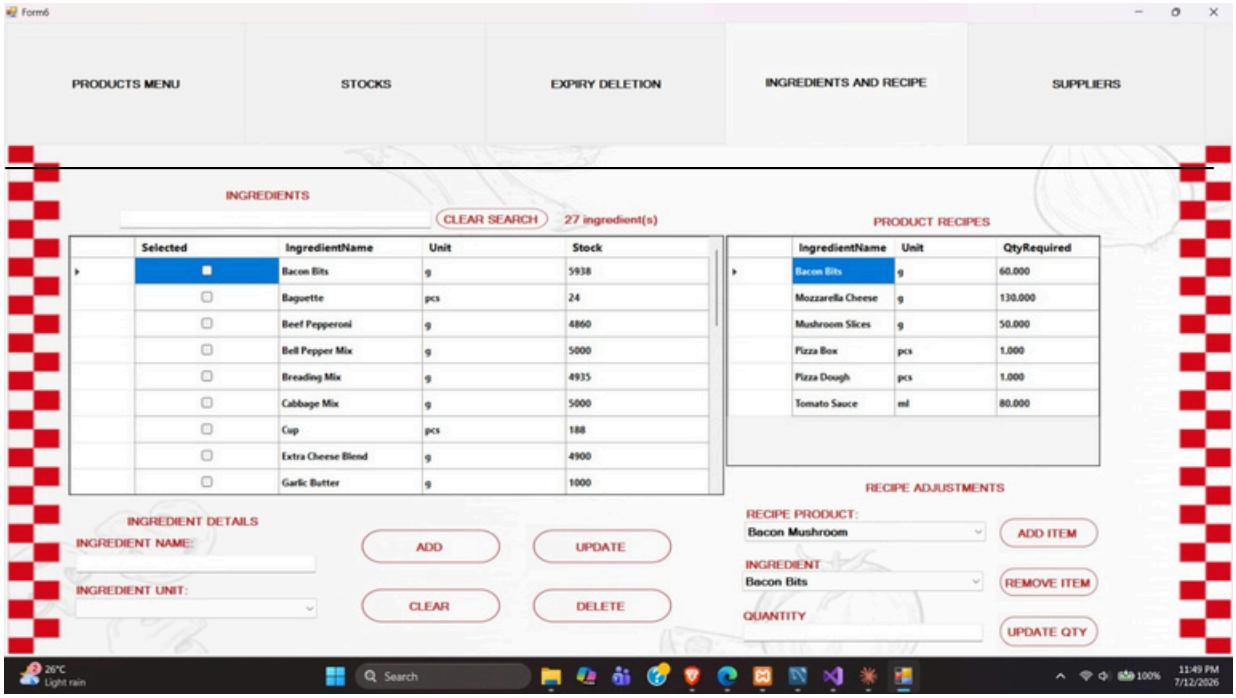


Figure 7. Product Inventory — Ingredients and Recipe tab

- Manage supplier records — add, update, and delete supplier name, contactperson, contact number, and address (Suppliers tab).

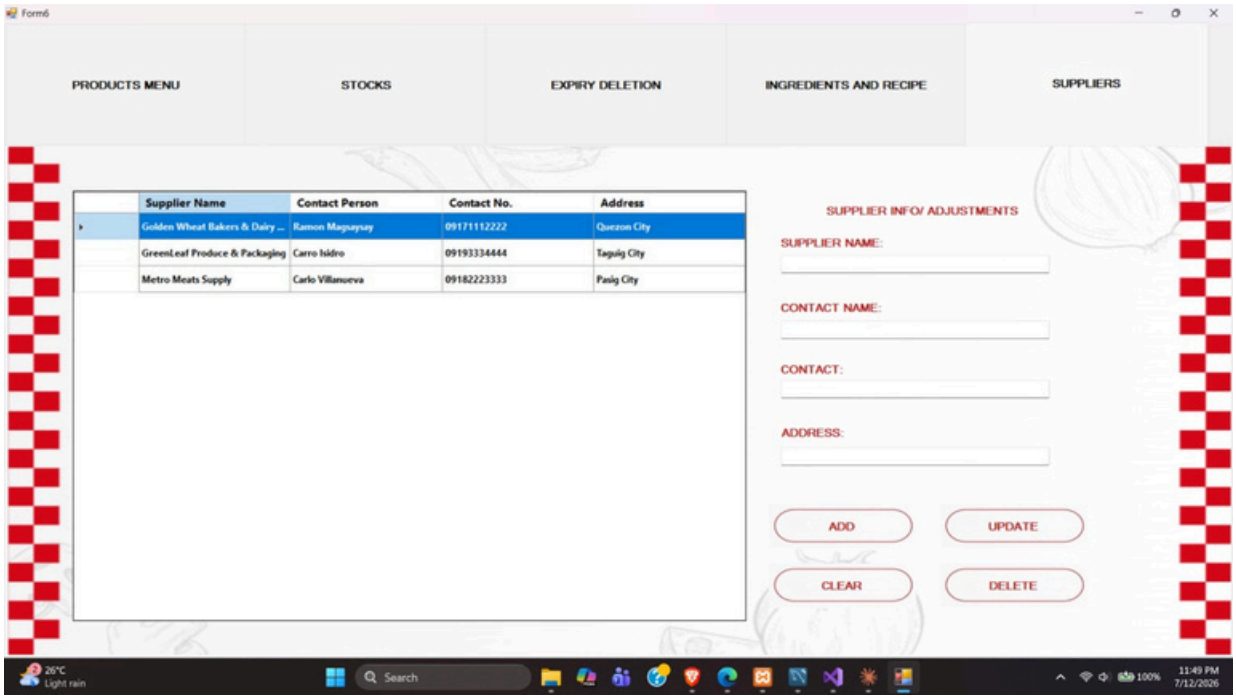


Figure 8. Product Inventory — Suppliers tab

- Process and manage sales transactions — filter by customer/date, create or void an order, edit line items and quantities on pending orders, confirm payment, and reprint receipts. Completed transactions are locked from further edits.

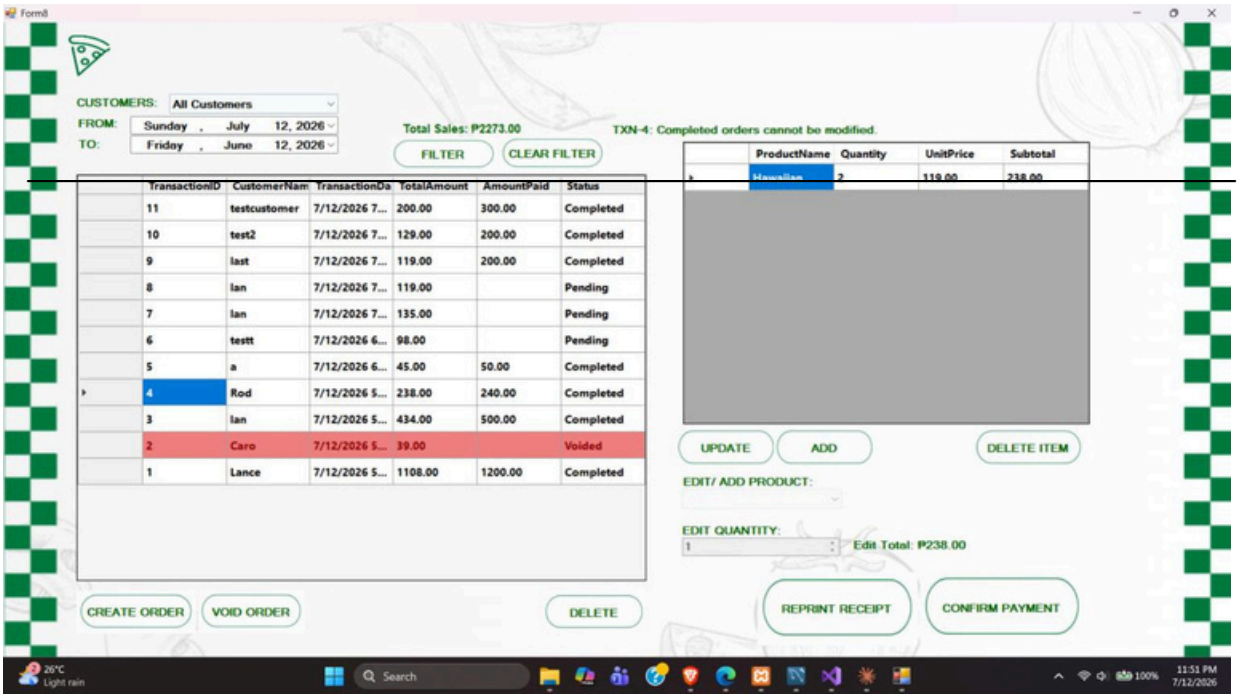


Figure 9. Transactions tab, with a completed transaction's item breakdown shown on the right

- Record and manage expenses — add, update, or delete expenses, filter by category and daterange. Restock expenses are generated automatically when ingredients are restocked and cannot be manuallyedited or deleted.

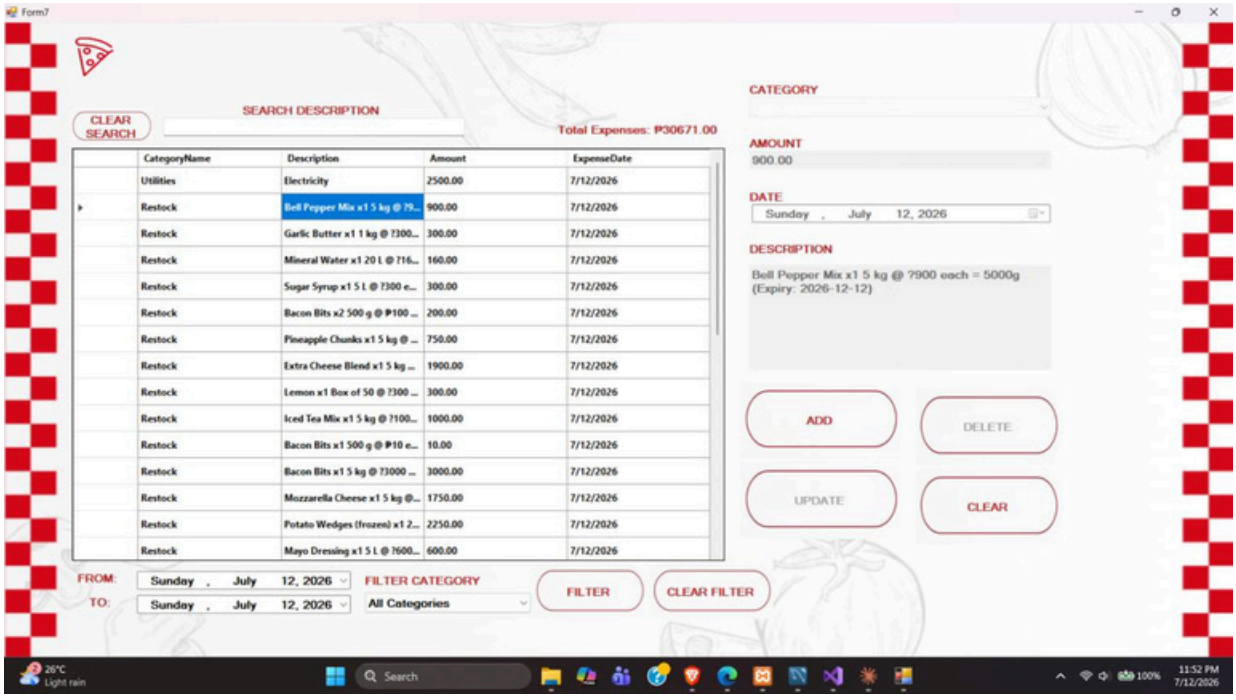


Figure 10. Expenses tab — viewing an auto-generated restock expense entry

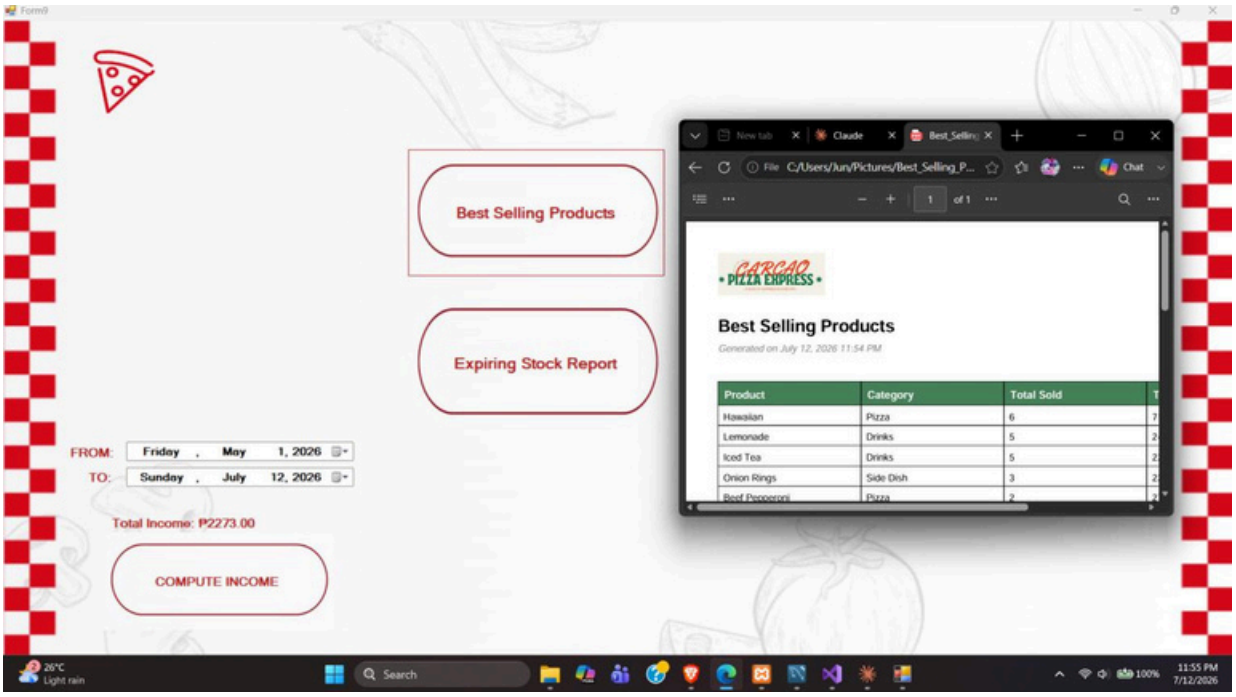


Figure 12. Reports tab, with a generated Best Selling Products PDF report

- Generate reports — Best Selling Products, Expiring Stock Report, and other sales/income reports filtered by date range, plus a Compute Income function for total income over a period.

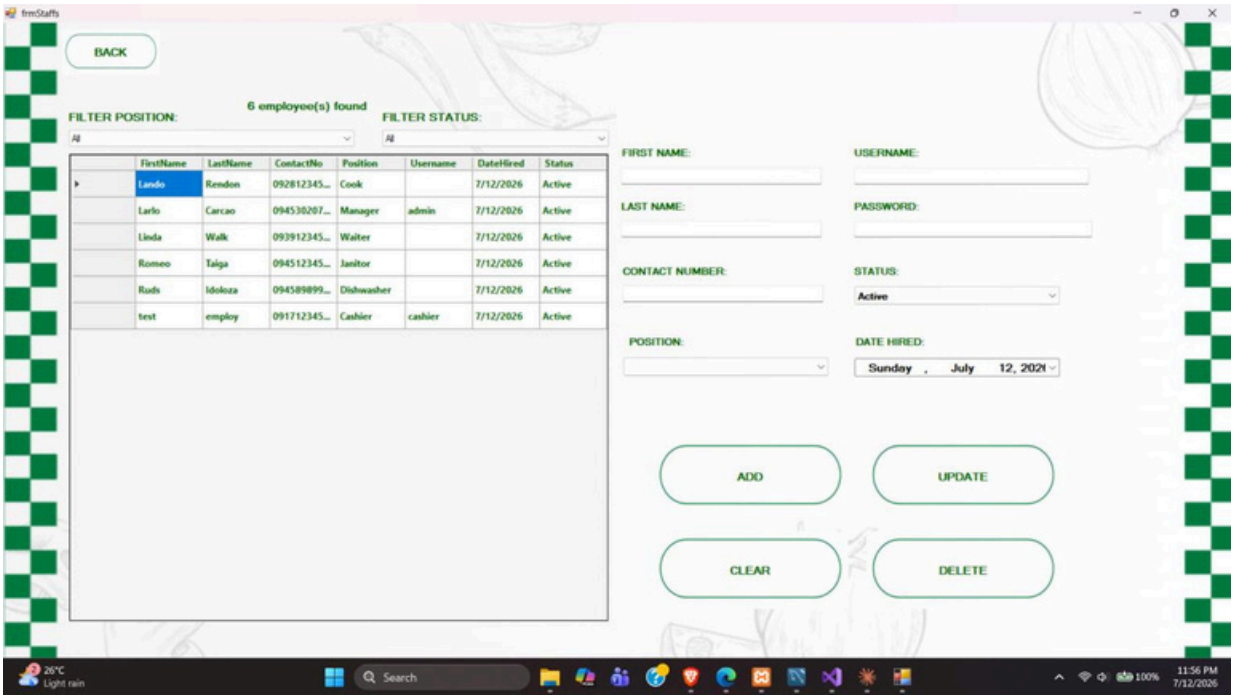


Figure 13. Employees tab (frmStaffs)

- Manage employee records — add, update, delete, and filter staff by position and status, including username/password for system access.

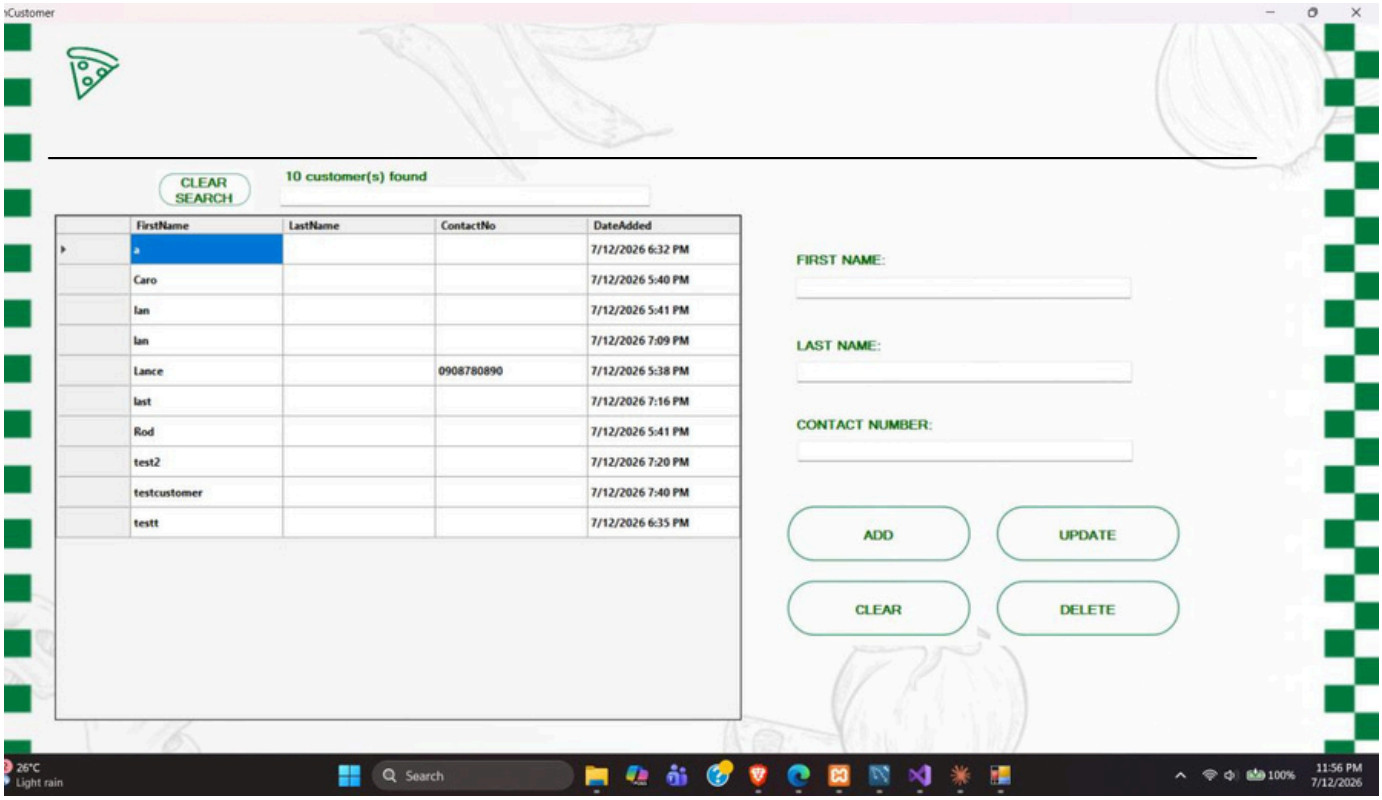


Figure 14. Customers tab

- Manage customer records — add, update, delete, and search customer information.

2. Cashier Functions (Limited Access)

Acashier logsinthroughthesame loginform butonly has access to the Transactions tab — the Product Inventory, Expenses, Reports, Employees, and Customers modules are restricted to the administrator.

- Process customer transactions — create new orders, edit pending order items and quantities, void orders, confirm payment, and print or reprint receipts.
- View transaction history and filter by customer or date range.

Order Placement

Ordersareplacedthrough the order screen, where items are browsed by category, added to the cart with quantity controls, and matchedwith the customer's name and optional contact details before checkout.

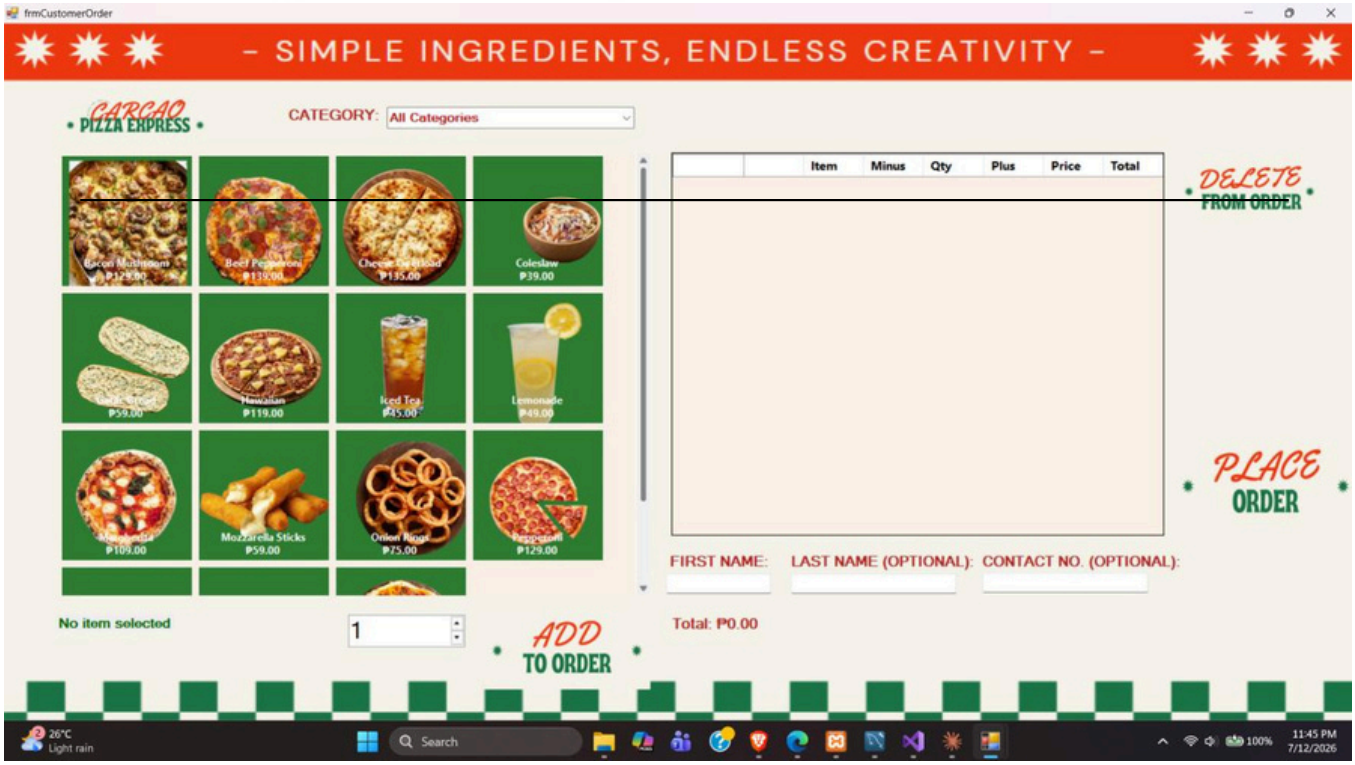


Figure 15. Customer order screen — menu, cart, and checkout details

Once an order is placed, the transaction is saved, its line items are recorded in transactiondetails, and the required ingredient quantities are deducted from the matching batches in ingredientbatch based on each product's recipe. When payment is confirmed, a receipt is generated and can be printed or reprinted at any time.

CONCLUSION

The CarcaoPizza Express POS Systemtiestogetherevery part of running the store — ordering, inventory, suppliers, staffing, andfinances — into one connected database. Because products are linked to recipes and recipes are linked to ingredientbatches, a single sale automatically keeps stock counts, expiry tracking, and cost records accurate without extramanual work. Separating administrator and cashier access also means front-of-house staff can process orders quicklywithout exposing sensitive data like recipes, supplier costs, or employee records.

Overall, thesystem reduces the manual effort of tracking sales, stock, and expenses separately. Reports and receipts are generateddirectly from the same live data used to run the store, giving management an accurate, real-time picture of salesperformance, ingredient usage, and income for better day-to-day decision-making.

Sample Outputs

The sequence belowtraces a single order from placement to payment, showing how the system's outputs connect end to end.

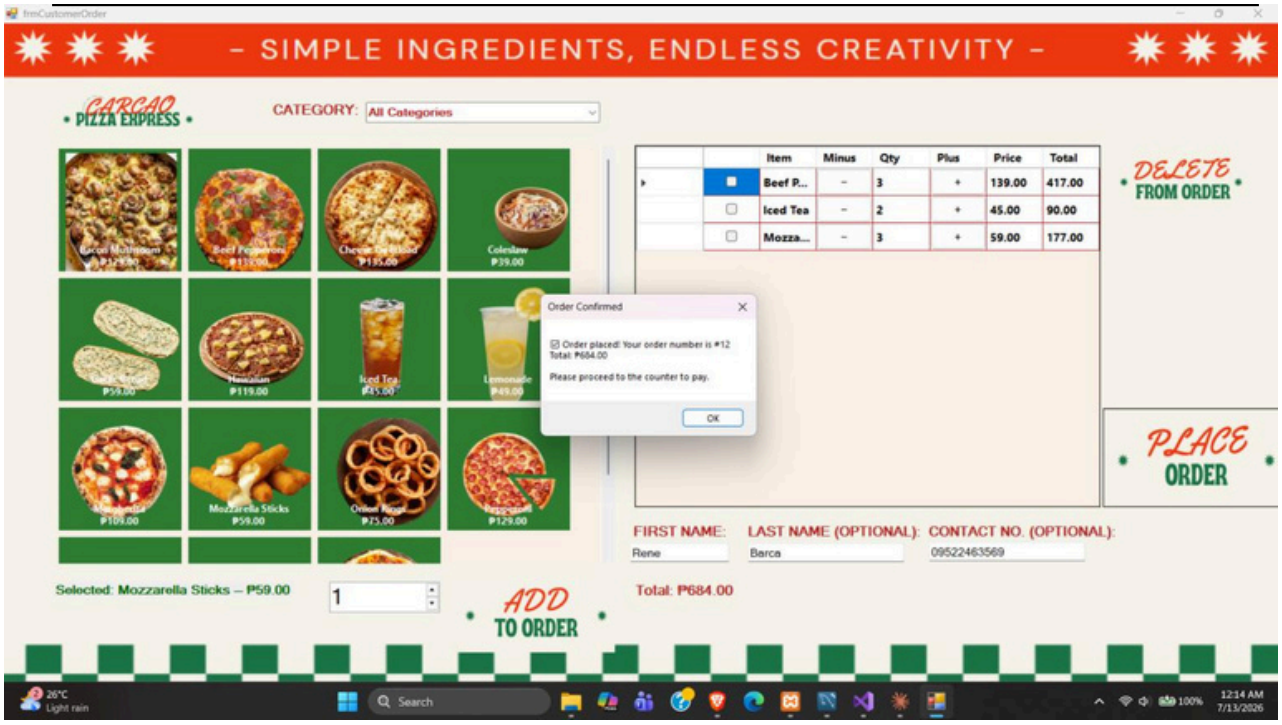


Figure 16. Order confirmation dialog after the customer places an order

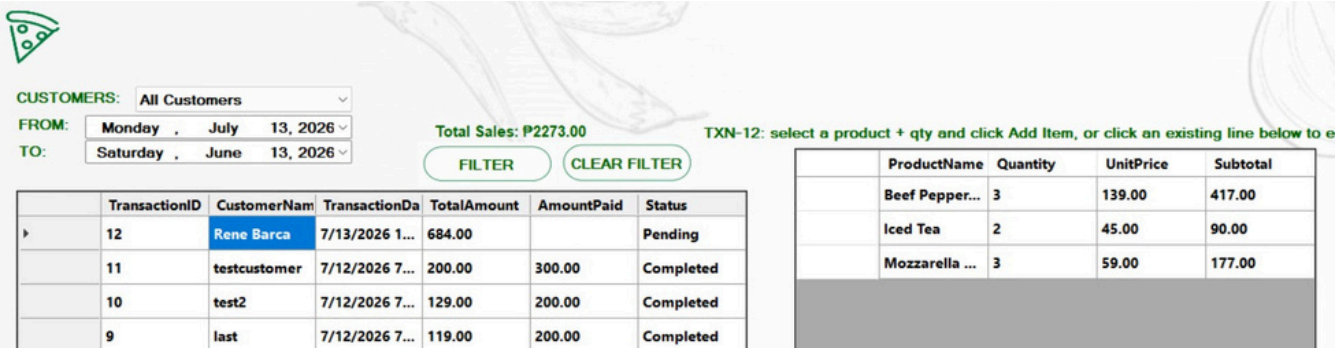


Figure 17. The same order appears in the Transactions tab with a Pending status, ready for payment

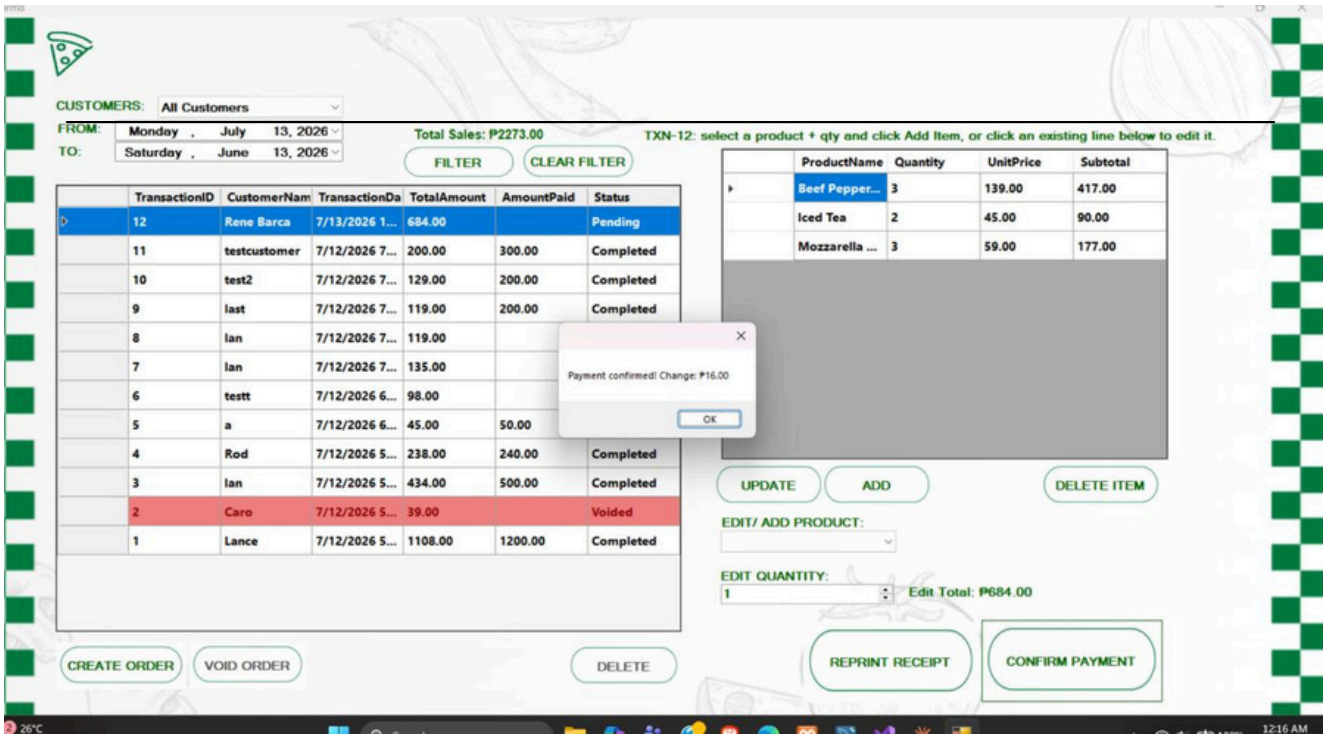


Figure 18. Payment confirmation dialog showing the computed change after Confirm Payment

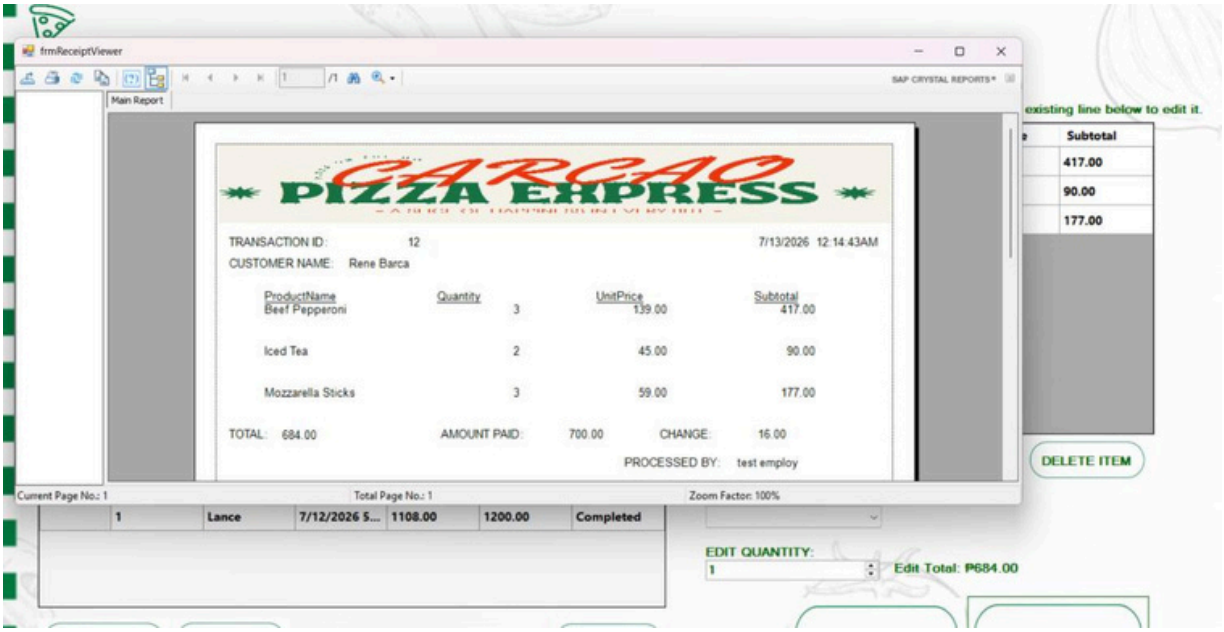


Figure 19. The final printed receipt for the completed transaction

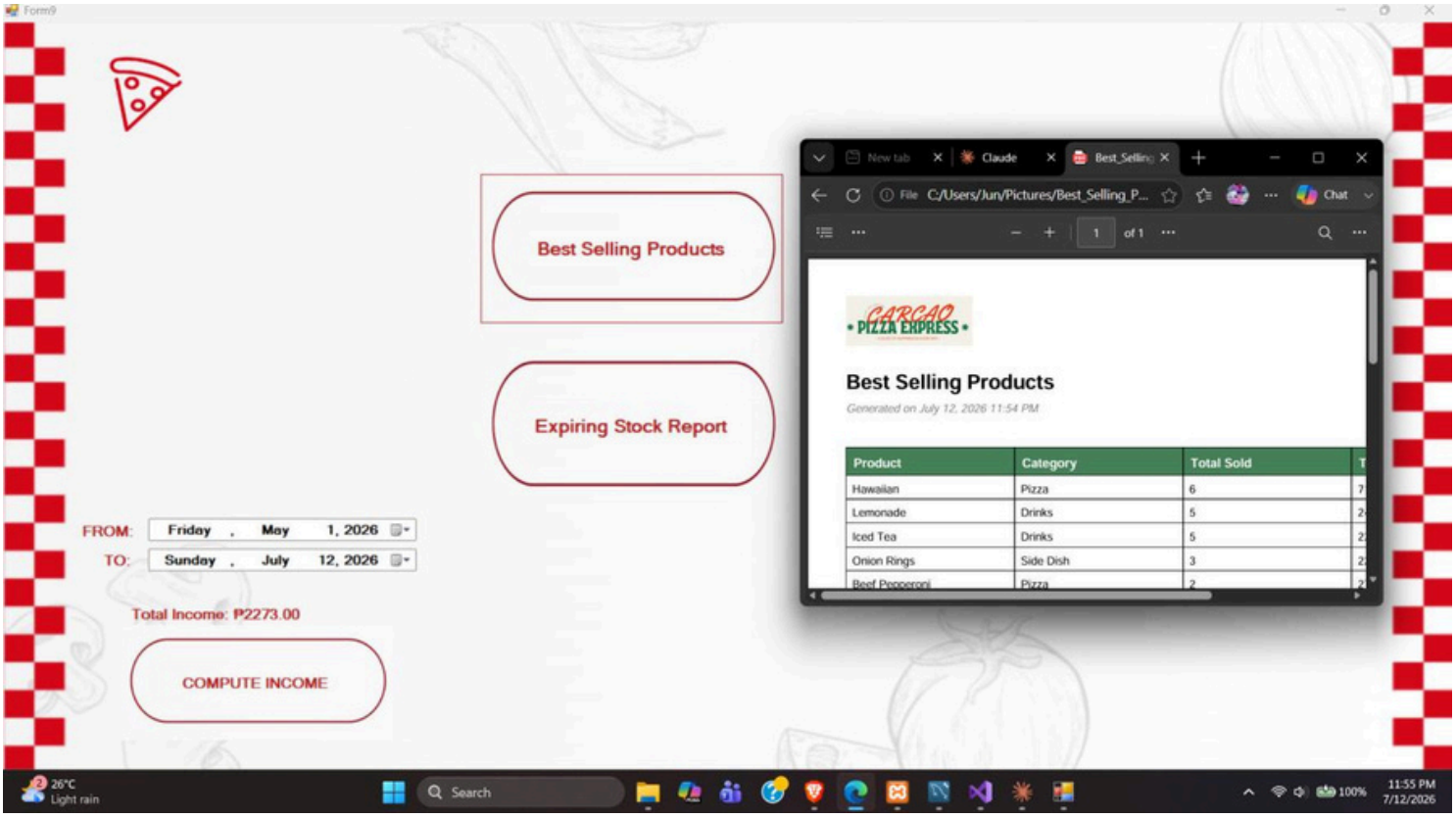


Figure 20. Sample generated report (Best Selling Products)